Oligopoly

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 18, Oligopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 18, "Oligopoly".

What you must know from this chapter

1. **Conditions:** what makes a market an oligopoly.
2. **Cartels:** firms acting together as one monopoly.
3. **Collusion:** agreements to restrict output or fix prices, and why they break down.
4. How the outcome under oligopoly compares with monopoly and perfect competition.

The model used in this lecture

Two firms, A and B, sell an identical product.

- Market demand is a straight line:

$$P = a - bQ, \qquad Q = q_A + q_B.$$

- q_A and q_B are the two firms' outputs. Q is total output.
- Both firms have the same constant marginal cost, c, and no fixed cost.
- So each unit costs c, and average cost is also c.

Part 1

What Makes a Market an Oligopoly?

Oligopoly

Definition: Oligopoly
A market with only a few sellers, each offering a product similar or identical to the others'.

- **Few sellers:** a handful of firms supply most of the market.
- **Similar or identical products.**
- **Barriers to entry:** new firms find it hard to enter.
- **Interdependence:** each firm's profit depends on what its rivals do.

Examples: commercial aircraft (Boeing and Airbus); U.S. wireless phone service.

Definition: Duopoly
An oligopoly with exactly two sellers.

Interdependence, written as an equation

Firm A's profit is its profit per unit times its output.

$$
\begin{aligned}
\pi_A &= (P - c) \times q_A && \text{profit per unit times units} \\
&= (a - bQ - c) \times q_A && \text{replace } P \text{ with } a - bQ \\
&= (a - b\,q_A - b\,q_B - c) \times q_A && \text{replace } Q \text{ with } q_A + q_B
\end{aligned}
$$

- Firm B's output, q_B, appears in firm A's profit.
- The more B sells, the lower the price, and the less A earns on each unit.

Finding

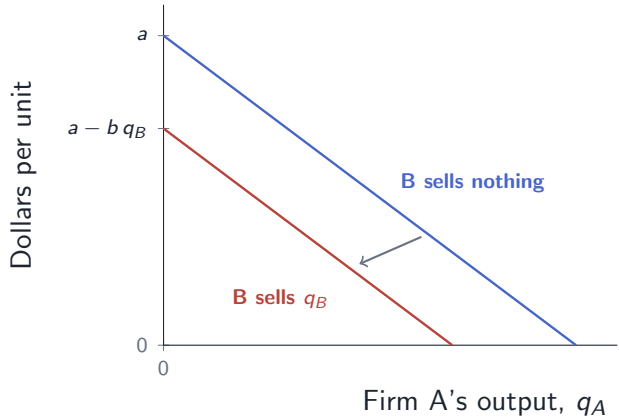
In an oligopoly, a firm's profit depends on its rival's choice. In perfect competition and in monopoly, no rival's choice enters a firm's profit.

The demand left for firm A

- Rearrange demand:

$$P = (a - b\,q_B) - b\,q_A.$$

- For a given q_B, this is the demand left for firm A.
- It starts at $a - b\,q_B$ and has slope $-b$.
- When B sells more, A's demand shifts down.



Four market structures

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Sellers	Many	Many	**A few**	One
Product	Identical	Differentiated	**Similar or identical**	No close substitutes
Entry	Free	Free	**Difficult**	Blocked
Rivals matter?	No	Little	**Yes**	No rivals

Finding

What sets oligopoly apart is interdependence: each firm must allow for what its rivals do.

Part 1 summary

- Oligopoly: a few sellers of similar or identical products.
- Entry is difficult.
- Interdependence: firm A's profit depends on firm B's output.
- When B sells more, the demand left for A shifts down.
- Compared with perfect competition and monopoly: there, no rival's choice enters a firm's profit.

Check: Part 1

Which of the following examples illustrates an oligopoly market?

A A farmers' market with many individuals selling sweet corn and tomatoes

B A city whose electrical service is provided by one electric co-operative

C A city with two firms that are licensed to sell school uniforms for the local schools

D A city with many independently owned hair styling salons

Check: Part 1 – answer

Which of the following examples illustrates an oligopoly market?

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C A city with two firms that are licensed to sell school uniforms for the local schools

D A city with many independently owned hair styling salons

Finding

Two sellers of the same product, with entry blocked by licensing: a duopoly. A is perfect competition, B is monopoly, D is monopolistic competition.

Part 2

Cartels: Acting as One Monopoly

Collusion and cartels

Definition: Collusion

An agreement among firms in a market about the quantities to produce or the prices to charge.

Definition: Cartel

A group of firms acting together. A successful cartel acts as one monopolist.

- The cartel chooses total output Q to maximize joint profit.
- Then it splits Q between the members.

The cartel's output: the monopoly rule

As one monopolist, the cartel produces where MR = MC.

$\text{MR} = a - 2bQ$	straight-line demand: MR twice as steep (Chapter 16)
$a - 2bQ = c$	set MR = MC
$2bQ = a - c$	add $2bQ$ to both sides; subtract c
$Q_{\text{cartel}} = \dfrac{a-c}{2b}$	divide both sides by $2b$

Price, from demand:

$$P_{\text{cartel}} = a - b \times \frac{a-c}{2b}$$ put Q_{cartel} into $P = a - bQ$

$$= a - \frac{a-c}{2} = \frac{a+c}{2}$$ the b's cancel

The benchmark: perfect competition

Under perfect competition, price equals marginal cost.

$$
\begin{aligned}
a - bQ &= c && \text{set } P = \text{MC} \\
Q_{\text{comp}} &= \frac{a-c}{b} && \text{solve for } Q \\
P_{\text{comp}} &= c
\end{aligned}
$$

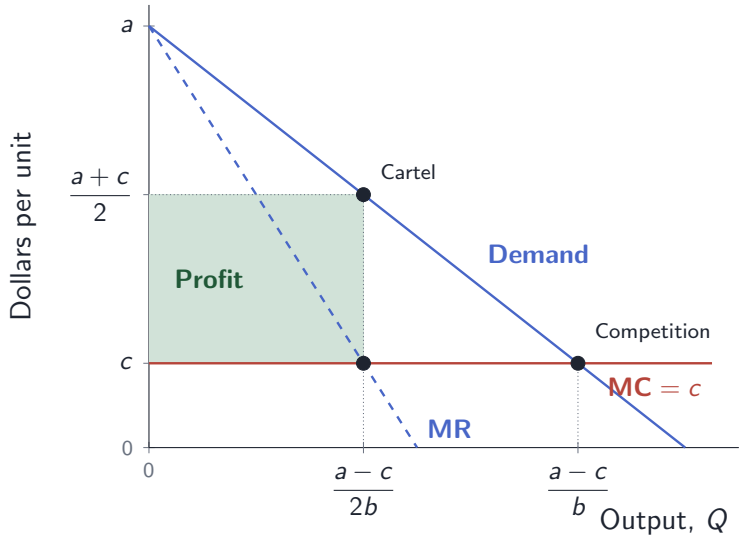
Compare the two outputs:

$$Q_{\text{cartel}} = \frac{a-c}{2b} = \frac{1}{2} \times \frac{a-c}{b} = \frac{1}{2} Q_{\text{comp}}.$$

Finding

A cartel produces half the competitive output and charges $\frac{a+c}{2}$, above marginal cost c.

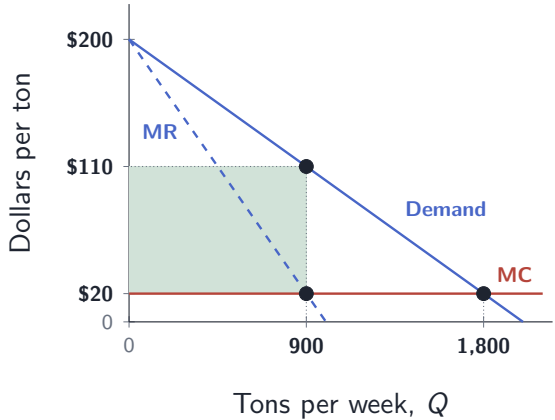
The cartel on the graph



Example: a cartel of two cement firms

$P = 200 - Q/10$, so $a = 200$, $b = 1/10$; $c = \$20$.

1. $Q_{\text{cartel}} = \dfrac{200 - 20}{2 \times 1/10} = 900$ tons.
2. $P_{\text{cartel}} = \dfrac{200 + 20}{2} = \$110$.
3. Profit $= (110 - 20) \times 900 = \$81{,}000$.
4. Split equally: 450 tons and \$40,500 each.
5. Competition: $Q = \dfrac{180}{1/10} = 1{,}800$ tons at \$20.



Part 2 summary

- Collusion: firms agree on quantities or prices.
- A cartel acts as one monopolist: MR = MC.
- $Q_{\text{cartel}} = \dfrac{a-c}{2b}$ and $P_{\text{cartel}} = \dfrac{a+c}{2}$.
- The members split the output and the profit.
- Compared with perfect competition: half the output, and a price above MC.

Check: Part 2

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

C price would rise and quantity would fall.

D price would fall and quantity would rise.

Check: Part 2 – answer

If duopoly firms that are not colluding were able to successfully collude, then

A price and quantity would rise.

B price and quantity would fall.

C price would rise and quantity would fall.

D price would fall and quantity would rise.

Finding

A cartel restricts output to the monopoly level, so quantity falls and price rises.

Part 3

Why Collusion Breaks Down

The question

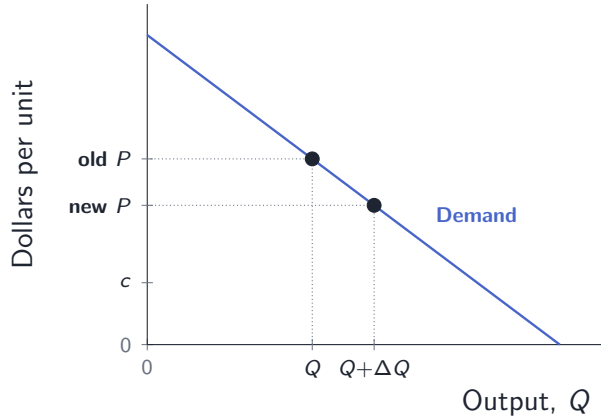
- The cartel output gives the two firms the most profit they can earn together.
- Yet cartels break down. Why would a member break an agreement that serves the pair best?
- To answer, follow what happens when one firm produces more than it agreed.
- We track three profits: the cartel's (both firms together), firm A's and firm B's.

Marginal cost in this model

- Each unit costs the same amount, c, and there is no fixed cost.
- So total cost is $\text{TC} = c \times Q$.
- One more unit adds exactly c to total cost. So marginal cost is $\text{MC} = c$, the same for every unit.
- A unit sold at price P therefore earns $P - c$ over its cost.

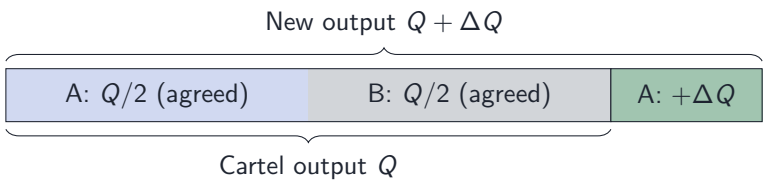
One product, one price

- The two firms sell the same product, so buyers pay the same price to either firm.
- The price depends on *total* output: $P = a - bQ$, with $Q = q_A + q_B$.
- If either firm sells more, Q rises and the price falls.
- The lower price applies to every unit sold, A's and B's alike.



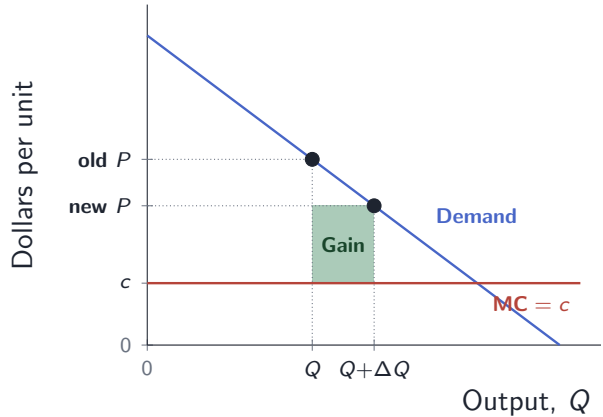
The agreement and the cheat

- **The agreement:** the cartel produces its output Q and each firm makes half, $Q/2$.
- **Cheating:** firm A produces ΔQ more than it agreed. Firm B keeps to its agreed $Q/2$.
- Total output rises from Q to $Q + \Delta Q$, so the price falls from old P to new P.



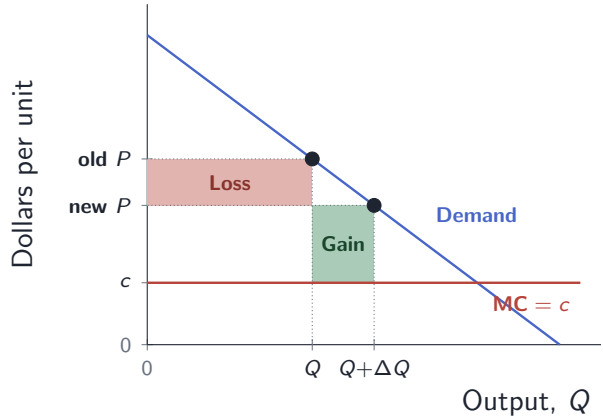
The gain: what A makes on the extra units

- A sells ΔQ extra units at the new price.
- Each extra unit earns new $P - c$ over its cost.
- So the gain is (new $P - c$) $\times$ ΔQ: the green rectangle.
- The gain goes to A alone: the extra units are A's.
- This is the *output effect*.



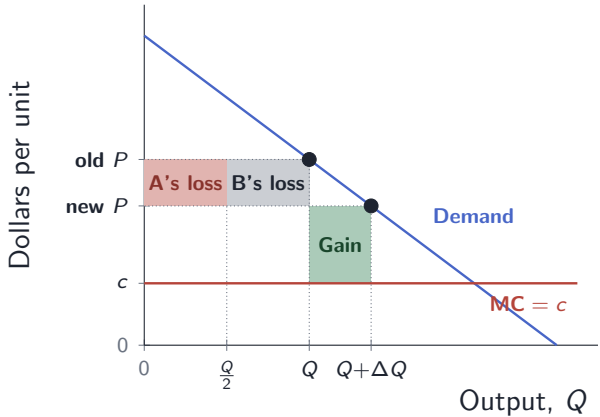
The loss: the lower price on units already sold

- Before A cheated, Q units were sold at the old price.
- Those same Q units now sell at the new, lower price.
- Each loses (old $P-$ new P). So the loss is (old P – new P) $\times$ Q: the pink rectangle.
- This is the *price effect*.



Who bears the loss?

- Of the Q units already sold, A sells half and B sells half.
- So the pink loss splits into two equal parts.
- A bears the loss on its own $Q/2$ units (pink).
- B bears the loss on its $Q/2$ units (grey).



Three profit changes

- **Cartel** (A and B together): gets the whole gain, bears the whole loss.

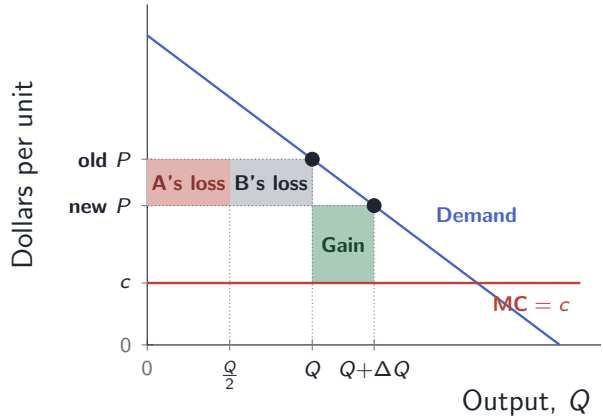
  Change = green − (pink + grey).

- **Firm A:** gets the whole gain, bears only its own loss.

  Change = green − pink.

- **Firm B:** gets none of the gain, bears its own loss.

  Change = − grey.



Why firm A cheats

- The cartel output is the best the pair can do. So producing more *lowers* the cartel's profit: the green is smaller than pink + grey.
- Compare A's change with the cartel's change. A's leaves out the grey:

$$\text{A's change} = \text{cartel's change} + \text{grey}.$$

- The grey is B's loss. A does not pay it, so A does not count it.
- So A can gain even though the cartel loses. It does whenever the green is bigger than A's own pink.

Finding

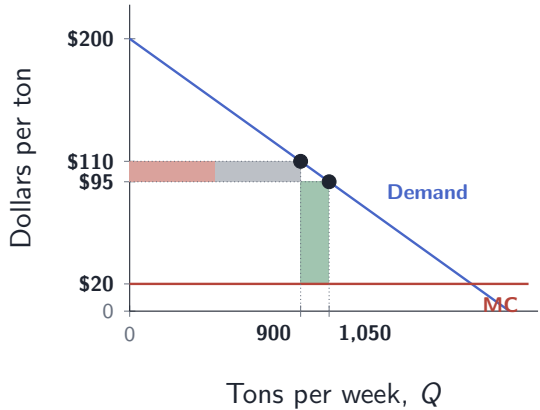
A cheater keeps all the gain from the extra output but pays only part of the loss. The rest falls on its partner. That is why cartels break down.

The cement cartel: the exact numbers

Agreement: 450 tons each, $P = \$110$. A makes 600; B keeps 450. Then $Q = 1{,}050$ and $P = \$95$.

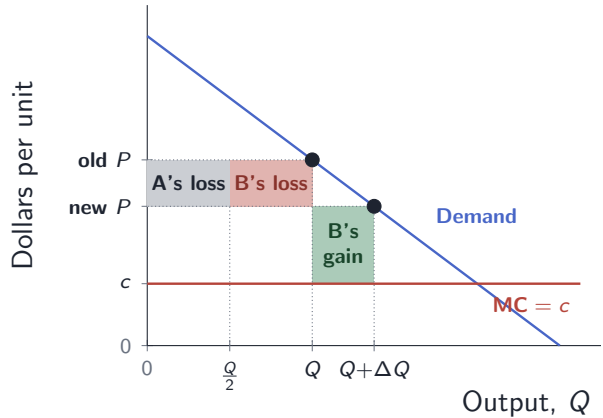
Green: $(95 - 20) \times 150$	\$11,250
Pink, A's 450 tons: 15×450	\$6,750
Grey, B's 450 tons: 15×450	\$6,750
Cartel: $11{,}250 - 13{,}500$	$-\$2{,}250$
Firm A: $11{,}250 - 6{,}750$	$+\$4{,}500$
Firm B	$-\$6{,}750$

Check: $+4{,}500 = -2{,}250 + 6{,}750$.



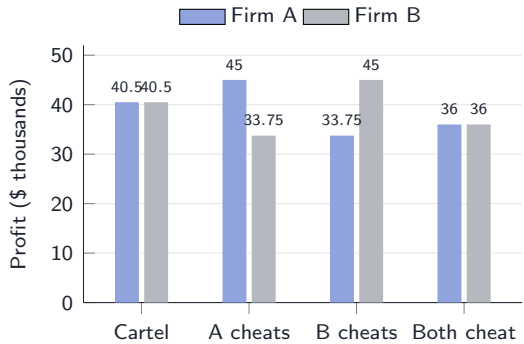
The same is true for firm B

- Now B produces ΔQ more and A keeps to its agreed $Q/2$.
- B gets the whole gain (green) and bears only its own loss.
- A bears the loss on its units and gets none of the gain.
- Cement: B gains \$4,500; A loses \$6,750.
- The firms are identical, so each sees the same temptation.



If both cheat, both lose

- Each firm is tempted, so both make 600 tons.
- $Q = 1{,}200$, so $P = \$80$.
- Each earns $(80 - 20) \times 600 = \$36{,}000$.
- That is less than the \$40,500 each earned in the cartel.



Finding

Each firm's temptation is real, but when both give in, both end up worse off than in the cartel.

The payoff table

Each firm chooses 450 tons (keep the agreement) or 600 tons (cheat).

	B makes 450	B makes 600
A makes 450	($40,500, $40,500)	($33,750, $45,000)
A makes 600	($45,000, $33,750)	($36,000, $36,000)

- **How to read it:** each cell is one pair of choices. The first number is A's profit; the second is B's.
- **For A, compare within a column** (B's choice held fixed). If B makes 450: 600 gives A $45,000, more than $40,500.
- If B makes 600: 600 gives A $36,000, more than $33,750.
- So 600 is better for A whatever B does. Comparing within a row shows the same for B.

Definition: Dominant strategy

A choice that is best whatever the other player does. Here: 600 tons, for both firms.

Nash equilibrium and the prisoners' dilemma

Definition: Nash equilibrium

A pair of choices in which each player's choice is its best, given the other's choice. Neither gains by changing its own choice alone.

- At (600, 600): if A alone switches to 450, its profit falls from \$36,000 to \$33,750. The same holds for B.
- So (600, 600) is the Nash equilibrium.
- Yet at (450, 450) both would earn \$40,500.

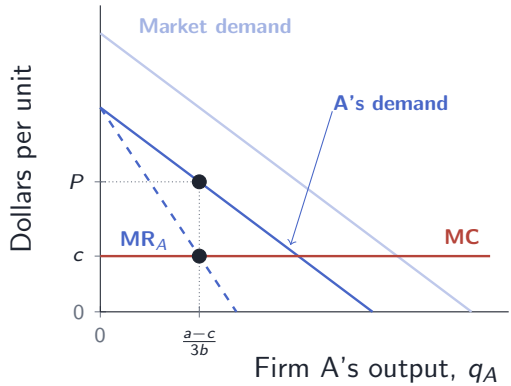
Definition: Prisoners' dilemma

A game in which each player's own best choice leaves both worse off than if they cooperated.

Where the firms end up

- Each firm takes its rival's output as given and sets its own $\text{MR} = \text{MC}$.
- Result: each firm produces $\dfrac{a-c}{3b}$.
- That is more than its cartel share, $\dfrac{a-c}{4b}$.
- Cement: 600 tons each, against 450 in the cartel.

Working: Appendix A.3.



Part 3 summary

- One product, one price: if either firm sells more, the price falls on every unit.
- A cheater gets the whole gain on its extra output (output effect).
- The loss on units already sold (price effect) is shared: each firm bears it on its own units.
- A's change = cartel's change + B's loss. So A can gain while the cartel loses.
- The same holds for B. Each cheats; both end up worse off.
- Dominant strategy: 600 tons. Nash equilibrium: (600, 600). A prisoners' dilemma.

Check: Part 3

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

B the number of firms is always large.

C costs to the firms in a cartel are continually rising.

D each firm has an incentive to deviate from its agreed output level.

Check: Part 3 – answer

Cartels are difficult to maintain because

A the monopoly output is very difficult to determine.

B the number of firms is always large.

C costs to the firms in a cartel are continually rising.

D each firm has an incentive to deviate from its agreed output level.

Finding

A cheater keeps the whole gain from extra output but bears only part of the loss, so each member wants to produce more.

Part 4

Market Outcomes Compared

Three outcomes from the same demand and cost

Structure	Output	Price	Total profit	P and MC
Monopoly or cartel	$\frac{a-c}{2b}$	$\frac{a+c}{2}$	Highest	$P > \text{MC}$
Oligopoly, two firms	$\frac{2(a-c)}{3b}$	$\frac{a+2c}{3}$	Lower	$P > \text{MC}$
Perfect competition	$\frac{a-c}{b}$	c	Zero	$P = \text{MC}$

The oligopoly price, from demand: $P = a - b \times \frac{2(a-c)}{3b} = a - \frac{2(a-c)}{3} = \frac{a+2c}{3}$.

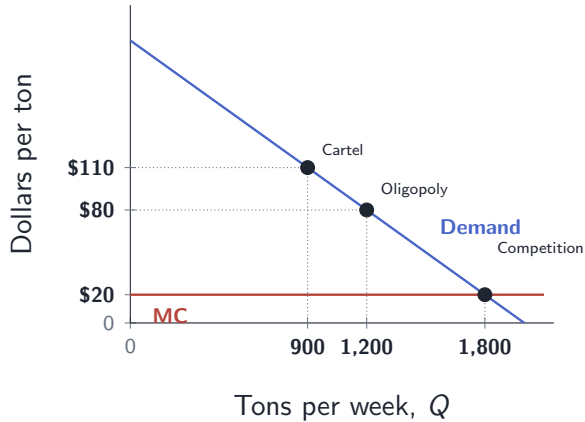
Finding

Oligopoly output lies between monopoly and perfect competition, and so does its price.

The cement market: three outcomes

	Q	P
Cartel	900	\$110
Oligopoly	1,200	\$80
Competition	1,800	\$20

Total profit: \$81,000, then
$2 \times 36{,}000 = \$72{,}000$, then \$0.



All four market structures in the long run

	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Price and MC	$P = \text{MC}$	$P > \text{MC}$	$P > \text{MC}$	$P > \text{MC}$
Long-run profit	Zero	Zero	Can be positive	Can be positive
Why	Free entry	Free entry	Entry is difficult	Entry is blocked
Output	Efficient	Below efficient scale	Below competitive	Lowest

Finding

Only perfect competition sets $P = \text{MC}$. In the other three, some buyers who value a unit above its cost do not get it.

Collusion and the law

- Collusion raises price and cuts output, so buyers lose.
- **Sherman Antitrust Act (1890):** agreements among competitors to fix prices or restrict output are illegal.
- **Clayton Act (1914):** individuals harmed by illegal agreements can sue to recover damages.
- **Example:** in 1996 Archer Daniels Midland agreed to plead guilty to fixing the prices of lysine and citric acid, two food and feed additives, and paid a $100 million fine.

Source: U.S. Department of Justice, press release, 15 October 1996.

Part 4 summary

- Cartel: output $\frac{a-c}{2b}$, the lowest; price $\frac{a+c}{2}$, the highest.
- Oligopoly with two firms: output $\frac{2(a-c)}{3b}$; price $\frac{a+2c}{3}$.
- Perfect competition: output $\frac{a-c}{b}$; price c.
- Oligopoly lies between monopoly and perfect competition.
- Price-fixing is illegal under the Sherman Act.

Check: Part 4

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Check: Part 4 – answer

The equilibrium quantity in markets characterized by oligopoly is

A higher than in monopoly markets and higher than in perfectly competitive markets.

B higher than in monopoly markets and lower than in perfectly competitive markets.

C lower than in monopoly markets and higher than in perfectly competitive markets.

D lower than in monopoly markets and lower than in perfectly competitive markets.

Finding

$\frac{a-c}{2b} < \frac{2(a-c)}{3b} < \frac{a-c}{b}$. In cement: $900 < 1{,}200 < 1{,}800$.

Which rule answers which question

Question	Rule
What defines an oligopoly?	A few sellers, similar products, difficult entry, interdependence.
What does a cartel produce?	The monopoly output: MR = MC, $Q = \frac{a-c}{2b}$.
Why does a cartel break down?	Each member gains by producing more: $\Delta\pi_A = -\,q_B\,\Delta P > 0$.
What is a dominant strategy?	The best choice whatever the rival does.
What is a Nash equilibrium?	Each choice is best given the other's choice.
Where does oligopoly output lie?	Between monopoly and perfect competition.
Is collusion legal?	No: price-fixing is illegal under the Sherman Act.

Appendix A.1: the three profit changes in symbols

Old price P_0, new price P_1, so $\Delta P = P_1 - P_0 < 0$. A adds ΔQ; each firm had $Q/2$.

$$\begin{aligned}
\text{Gain (green)} &= (P_1 - c)\,\Delta Q \\
\text{Loss on all } Q \text{ units} &= -\,Q\,\Delta P, \quad \text{split } -\tfrac{Q}{2}\Delta P \text{ (A) and } -\tfrac{Q}{2}\Delta P \text{ (B)} \\
\Delta\pi_{\text{cartel}} &= (P_1 - c)\,\Delta Q + Q\,\Delta P \\
\Delta\pi_A &= (P_1 - c)\,\Delta Q + \tfrac{Q}{2}\,\Delta P \\
\Delta\pi_B &= \tfrac{Q}{2}\,\Delta P \\
\Delta\pi_A &= \Delta\pi_{\text{cartel}} - \Delta\pi_B = \Delta\pi_{\text{cartel}} + \text{B's loss}
\end{aligned}$$

At the cartel output, $\Delta\pi_{\text{cartel}} < 0$ for any increase; A gains whenever B's loss exceeds that fall.

Appendix A.2: the cement arithmetic

Demand $P = 200 - Q/10$; $c = \$20$; agreed output 450 tons each.

- Cartel: $Q = 900$, $P = 200 - 90 = \$110$; each earns $(110 - 20) \times 450 = \$40{,}500$; together \$81,000.
- A cheats: $Q = 600 + 450 = 1{,}050$, $P = 200 - 105 = \$95$. The price falls by \$15.
- Green: $(95 - 20) \times 150 = \$11{,}250$. Pink: $15 \times 450 = \$6{,}750$. Grey: $15 \times 450 = \$6{,}750$.
- A: $(95 - 20) \times 600 = \$45{,}000 = 40{,}500 + 11{,}250 - 6{,}750$.
- B: $(95 - 20) \times 450 = \$33{,}750 = 40{,}500 - 6{,}750$.
- Cartel: $45{,}000 + 33{,}750 = \$78{,}750 = 81{,}000 - 2{,}250$.
- Both cheat: $Q = 1{,}200$, $P = \$80$; each earns $(80 - 20) \times 600 = \$36{,}000$.

Appendix A.3: where the firms end up

Firm A takes B's output q_B as given.

$P = (a - b\,q_B) - b\,q_A$	the demand left for A: a straight line, slope $-b$
$\text{MR}_A = (a - b\,q_B) - 2b\,q_A$	MR is twice as steep (Chapter 16)
$c = a - b\,q_B - 2b\,q_A$	set $\text{MR}_A = \text{MC}$
$c = a - 3bq$	equal outputs, $q_A = q_B = q$
$q = \dfrac{a-c}{3b}$	solve for q

Cartel share: half of $\frac{a-c}{2b}$, that is $\frac{a-c}{4b}$. Cement: $q = \frac{180}{3/10} = 600$ tons; cartel share 450 tons.